

NeSS Mini Project

THERMISTORS

(Analysis and Applications)

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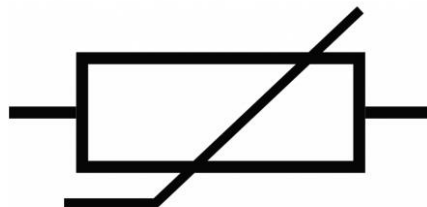
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Introduction

Thermistors are temperature-dependent variable resistors used in circuits for applications where temperature is a role-playing factor. Some examples are cooling systems, where they find direct usage in the cooling circuitry to adjust according to the temperature. In fact, every resistor is temperature-variable, but a thermistor is specially designed to be temperature-sensitive. A thermistor is basically a semiconductor metal oxide bead coated (generally) in epoxy glass. Thermistors, unlike resistors, are non-linear devices, i.e., there is no linear relationship between current through them and voltage across them. In this document we analyse their properties and explore their applications.

Equivalent Circuit

A thermistor is represented via a slashed line across a resistor, much like a rheostat:



For analysis, we will simply connect it to a Digital Multimeter (DMM) in resistance mode, and note its resistance values as they change with temperature change. The DMM sends a constant current through the thermistor, and calculates the potential drop across it, thus getting the resistance.

Types of Thermistors

These heat-dependent resistors can operate in one of two ways, either by increasing or decreasing their resistive value with changes in temperature. On this basis, there are two types of thermistors:

- Negative Temperature Coefficient of resistance (NTC)
- Positive Temperature Coefficient of resistance (PTC)

NTC thermistors decrease their resistive value as the operating temperature around them increases, and vice versa for PTC thermistors. Even small changes in temperature produce large responses in a thermistor, the response being negative for NTC and positive for PTC. This makes them ideal for accurate temperature measurement and control.

Thermistor Equation

An important characteristic of a thermistor is its ‘B’ value, which is a material constant of the ceramic material that makes the thermistor. This value ‘B’ describes the gradient of the R vs. T curve over a particular temperature range between two temperature points. This value is available in a thermistor’s datasheet provided by the manufacturer.

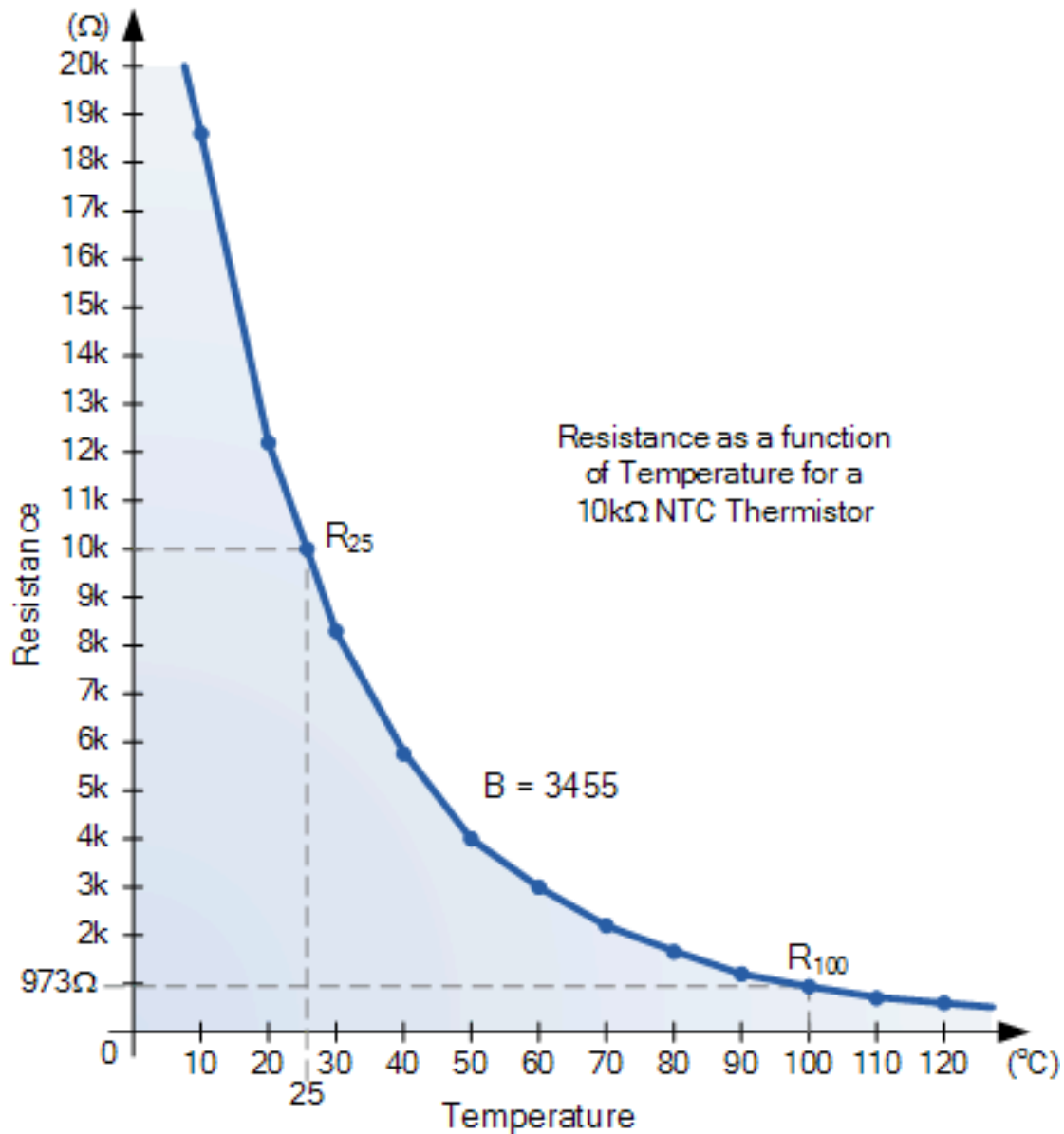
$$B_{T_1/T_2} = \frac{T_2 \times T_1}{T_2 - T_1} \times \ln\left(\frac{R_1}{R_2}\right)$$

where, T_1 is the first temperature point in Kelvin
 T_2 is the second temperature point in Kelvin
 R_1 is the thermistor’s resistance at T_1 in Ohms
 R_2 is the thermistor’s resistance at T_2 in Ohms

The base resistance R_1 at T_1 is generally specified in the specs, so that R_2 can be calculated for any temperature $T_2 > T_1$ in range.

NTC Thermistor Characteristic Curve

Here is the R/T curve for a 10 k Ω NTC thermistor with a B-value of 3455 between 25°C and 100°C. (10 k Ω is the base resistance)

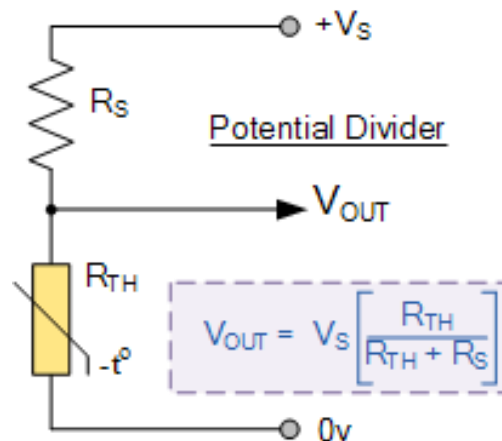


Source: <https://www.electronics-tutorials.ws/io/thermistors.html>

The resistance changes exponentially with changes in temperature, and so the characteristic curve is non-linear.

Applications of Thermistors

One obvious application of thermistors is temperature measurement. Since it is a resistive device, it can be directly employed in a circuit and translated into a suitable voltage output. A potential divider with one of the resistances as a thermistor may do the job.



Source: <https://www.electronics-tutorials.ws/io/thermistors.html>

However, the problem with passing current through a thermistor in this way is what is known as the self-heating effect: the power loss (IR^2) could be high enough to create more heat than can be dissipated by the thermistor, which affects the ambient temperature, giving false results. The increased temperature reduces its resistance (NTC), causing more current to pass through, further increasing the power loss and thence the heat dissipated, increasing the temperature further, all of which eventually causes a *Thermal Runaway*. All this can be evaded if the current through the thermistor is ensured to be below a safe value, by choosing a suitable value of R_S .

Thermistors are, in an extension to the above application, used in cooling systems where temperature feedback is read by it and fed to the cooling circuit, which accordingly changes the fan speed or coolant compression, depending on the type. In a simpler version, the decreased resistance on increasing temperature runs the fan motor faster, just like a manually controlled regulator of ceiling fans (rheostat).